Zexu Chen

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Research Interests

Computational materials science of structural and functional alloys; phase-field modeling of martensitic transformation and precipitation in NiTi shape memory alloys; GPU-accelerated high-performance computing; machine learning and Bayesian inference for materials property prediction and inverse alloy design.

Education

The Ohio State University, Columbus, OH Aug 2022 – May 2028 (expected)

Ph.D., Materials Science and Engineering

Advisor: Prof. Yunzhi Wang

The Ohio State University, Columbus, OH Conferred May 2025

M.S., Materials Science and Engineering (earned en route to Ph.D.)

Relevant coursework: Structure and Defects in Materials; Diffusion and Interface Kinetics; Thermodynamics of Materials.

Sichuan University / The Ohio State University Aug 2017 – May 2022

B.Sc., Materials Science and Engineering

Publications

Peer-reviewed journal articles

1. Sriram H., Feng L., **Chen Z.**, Wang Y. (2026). Soft and Hard Confinement Effects on Martensitic Transformation in NiTi Shape Memory Alloys: Insights from Phase-Field Simulations. *Shape Memory and Superelasticity.* https://doi.org/10.1007/s40830-026-00619-3
2. Abe O.O., Qiu Z., **Chen Z.**, Jinschek J.R., Gouma P.-I. (2021). Effect of Crystallite Size on the Low-Temperature Solid-Solid Phase Transformations in the WO_3 System. *Ceramics International* 47, 33476–33482. https://doi.org/10.1016/j.ceramint.2021.08.254

Manuscripts under review

- **Chen Z.**, Sriram H., Yu T., Feng L., Zhu J., Rajagopalan J., Wang Y. (2026). Programming Martensitic Transformation and Mechanical Response in NiTi Shape Memory Alloys through One-Dimensional Compositional Modulations. Under review, *Acta Materialia* (submitted July 2026); preprint available at SSRN 7302838. https://doi.org/10.2139/ssrn.7302838

Manuscripts in preparation

- **Chen Z.**, Wang Y. Coupling Grain Structure and Compositional Gradients to Control Variant Selection in Polycrystalline NiTi (working title). *In preparation.*
- **Chen Z.**, Hu K., Tan R., Zhang T., Wang Y. AI-Enabled Inverse Design of Superelastic Materials through Phase-Field Simulations and Machine Learning (working title). *In preparation.*

Research Experience

Graduate Researcher, Wang Group, The Ohio State University *Aug 2022 – Present*

Supported by the U.S. National Science Foundation, Award No. DMR-2333551 (PI: Prof. Yunzhi Wang).

Project 1 (ongoing, May 2026 – Present): Phase-Field Modeling of Grain-Boundary Concentration Modulations and Martensitic Transformation in Polycrystalline NiTi Shape Memory Alloys

- Advisor: Prof. Yunzhi Wang. Supported by the U.S. National Science Foundation, Award No. DMR-2333551.

Project 2 (ongoing, Nov 2025 – Present): Multiscale Modeling of Shear Band Control in Layered Metallic Glass and Crystalline Composites

- Combining phase-field modeling and shear transformation zone kinetic Monte Carlo simulations, in collaboration with Prof. Jinwoo Hwang (The Ohio State University), Prof. Yue Fan and Muchen Wang (University of Michigan), and Prof. Jagannathan Rajagopalan (Arizona State University) on experimental work. Supported by the U.S. National Science Foundation, Award No. CMMI-2611448 (PI: Prof. Yunzhi Wang).

Project 3 (ongoing, Jul 2025 – Present): Machine-Learning-Accelerated Inverse Design of Compositionally Graded Shape Memory Alloys with Experimental Validation

- In collaboration with Prof. Tianlong Zhang's group (Hong Kong University of Science and Technology) on the machine learning framework and Prof. Jagannathan Rajagopalan's group (Arizona State University) on experimental validation. Supported by the U.S. National Science Foundation, Award No. DMR-2333551.
- Developing Gaussian Process Regression surrogates that map 1D concentration-modulation parameters to transformation temperature and stress-strain behavior, enabling accelerated inverse design of NiTi alloy compositions.
- Generating training data from GPU phase-field campaigns and benchmarking kernel choices, LASSO feature selection, and Bayesian optimization for sampling the modulation design space.

Project 4 (lead author): Programming Mechanical Response of NiTi SMA via 1D Nanoscale Concentration Modulations

- Designed and ran 3D phase-field simulations of nanoscale concentration modulations to engineer linear superelastic stress-strain response of NiTi shape memory alloys for biomedical and actuator applications.
- Investigated habit-plane and twinning-plane formation in B19' martensite using martensite crystallography and Khachaturyan microelasticity theory; correlated microstructural descriptors with macroscopic mechanical behavior.

Project 5: Taming Martensitic Transformation via Dissolution of Precipitates in NiTi SMA

- Modeled Ni_4Ti_3 precipitation and B2-to-B19' martensitic transformation via multiphase-field simulations using Cahn-Hilliard and time-dependent Ginzburg-Landau evolution equations with explicit nucleation algorithm.
- Quantified effects of dissolution-induced concentration fields on transformation hysteresis and mechanical stability; produced parameter guidelines for collaborating experimentalists.
- Implemented GPU-accelerated phase-field code in C with OpenACC on the Ohio Supercomputer Center; managed large parametric job sweeps via SLURM for composition and load studies.

Earlier project (co-author): Crystallite-Size Effects on Phase Transformations in WO_3

- Synthesized WO_3 nanocrystals via sol-gel method; quantified crystallite size distributions from TEM micrographs to correlate microstructure with low-temperature solid-solid phase transformation behavior.

Team Member, "ML for Superconductivity Prediction", The Ohio State University *Spring 2026*

MATSCEN 7193.02 (Machine Learning for Materials Research), team project (4 members)

- Built XGBoost and GPR pipeline to predict superconducting parameters (λ, T_c) from Eliashberg spectral functions $\alpha^2F(\omega)$ on the Gibson *et al.* (*npj Comput Mater*, 2026) dataset of 806 DFT-validated superconductors; designed physics-informed features (frequency-weighted α^2F/ω bins) aligned with the Eliashberg integral.
- Tuned GPR with Matérn ($\nu = 1.5, 2.5$) + White kernels to capture spectral discontinuities; benchmarked spectral bin resolution (50, 200, 800) against generalization to characterize the curse-of-dimensionality trade-off, identifying physics-informed binning as a more robust route than naive high-resolution sampling.

Teaching Experience

Graduate Teaching Assistant, The Ohio State University *Oct 2023 – Oct 2024*

- Instructed two undergraduate courses: *Introduction to Engineering Materials* and *Modeling and Simulation Lab II*.
- Led weekly recitations, held office hours, and graded assignments and exams for approximately 40 students per course.

Mentoring

Mentor, Undergraduate Senior Capstone Design Team *Jan 2026 – Present*

Department of Materials Science and Engineering, The Ohio State University

- The team works on shape memory alloy design using phase-field simulations and machine learning.

Conference Presentations

- **TMS 2027** (Mar 2027). Oral (accepted): *Dilute Compositional Modulation as a Design Lever for Martensitic Transformation in NiTi Shape Memory Alloys.* **Chen Z.**, Sriram H., Feng L., Yu T., Zhu J., Rajagopalan J., Wang Y.
- **TMS 2027** (Mar 2027). Oral (accepted): *Coupling Grain Structure and Compositional Gradients to Control Variant Selection in Polycrystalline NiTi.* **Chen Z.**, Wang Y.
- **TMS 2027** (Mar 2027). Oral (accepted): *AI-Enabled Inverse Design of Superelastic Materials through Phase-Field Simulations and Machine Learning.* **Chen Z.**, Hu K., Tan R., Zhang T., Wang Y.
- **TMS 2026** (Mar 2026). Poster: *Tailoring Superelastic Behavior in NiTi SMAs via 1D Compositional Modulation: A Calibrated Phase-Field Framework.* **Chen Z.**, Wang Y.
- **MS&T 2025** (Oct 2025). Oral: *Tailoring Mechanical Properties of NiTi Shape Memory Alloys Through Key Parameters in One-Dimensional Compositional Modulation.* **Chen Z.**, Wang Y.
- **TMS 2025** (Mar 2025). Poster: *Regulating Stress-Strain Behavior of NiTi Shape Memory Alloys via 1D Nanoscale Concentration Modulations.* **Chen Z.**, Sriram H., Feng L., Wang Y.
- **MS&T 2023** (Oct 2023). Oral: *Fine-tuning Superelastic Behavior of NiTi SMAs via Nanoscale Concentration Modulation Created by Ni_4Ti_3 Nanoprecipitate Dissolution.* **Chen Z.**, Sriram H., Feng

L., Wang Y.

Professional Service

Reviewer, *Shape Memory and Superelasticity* (ASM International / Springer) *2026 – Present*

Reviewer, *Metallurgical and Materials Transactions A* (TMS / ASM International / Springer) *2026 – Present*

Technical Skills

Programming: Python (scikit-learn, PyTorch), C, MATLAB, Fortran, Bash
HPC & GPU: GPU acceleration (CUDA, OpenACC), SLURM, Linux, Git, Ohio Supercomputer Center
Simulation: Phase Field Modeling, Shear Transformation Zone Kinetic Monte Carlo, CALPHAD (Thermo-Calc), Molecular Dynamics, Constitutive Modeling
Machine Learning: Gaussian Process Regression, Bayesian optimization, XGBoost, LASSO regression, Convolutional Neural Networks, Generative Models (Conditional Variational Autoencoders, VAE-GANs), Physics-Informed Neural Networks (PINNs)
Data & Visualization: OVITO, ParaView, PyVista, JupyterLab, MATLAB Advanced Plotting
Domain Expertise: Shape memory alloys, martensitic transformation, ICME, materials informatics
Languages: English (Fluent), Mandarin (Native)

References available upon request.